Department of Microsystems Engineering

Intuition, Events and Probabilities

PD Dr. Cristian Pasluosta

Laboratory for Biomedical Microtechnology

UNI FREIBURG

Room: 201-00-022
Phone: 7251
Email: cristian.pasluosta@imtek.de

Topics of the day:

Set Theory

- History and applications
- Definitions: Set, Sub-Set, Tupel, Element, Trial
- Calculation rules I: Algebraic rules
- Calculation rules II: Kolmogorov Axioms (1-3)

Probability

- Definitions: Probability P, probability function p, Partition, probability measure
- Laplace probability
- Odds and Ratio (Ratio-Checker)
- Distributions: Laplace, discrete uniform distribution

Combinatorics

- Ordered/unordered samples with/without replacement

Examples for a set

A throw of the die has the possible results: 1,2,3,4,5,6



For the throw of a die: Ω = {1,2,3,4,5,6}

The draw of a card from a deck of cards:



Ω = { ♦7, ♦ 8, ♦ 9, ♦ 10, ♦ J, ♦ Q, ♦ K, ♦ A,
♥7, ♥ 8, ♥ 9, ♥ 10, ♥ J, ♥ Q, ♥ K, ♥ A,
♣7, ♣ 8, ♣ 9, ♣ 10, ♣ J, ♣ Q, ♣ K, ♣ A,
♠7, ♠ 8, ♠ 9, ♠ 10, ♠ J, ♠ Q, ♠ K, ♠ A }

or:

Ω := {1,2,3,.....,30,31,32} (!)

Probabilities, Kolmogorov Axioms

A.N. Kolmogorov set up the following three axioms in (1933).



Andrey Nicolaevich Kolmogorov (1903-1987)

For a finite probability space, let Ω *be* $(\Omega \neq \varnothing)$, *and P be a function* $(\Omega \text{ a } ¡)$ *on the subsets of* Ω.
The function P holds the following properties :

i) $P(A) \geq 0$, *for* $A \subset \Omega$ (*Nichtnegativität*; *non – negativity*)

ii) $P(\Omega) = 1$ (*Normiertheit*; *normalization*)

iii) $P(A+B) = P(A \cup B) = P(A) + P(B)$, *for* $A \cap B = \varnothing$ (*Additivität*; *additivity*)

Implications on Kolmogorovs Axioms

The following implications can be considered the 101 of probability calculation (Einmaleins der Wahrscheinlichkeitsrechnung)

a) $P(\varnothing) = 0$

b) $0 \leq P(A) \leq 1$

c) $P(\overline{A}) = 1 - P(A)$, (*complementary subset*)

d) *From* $A \subset B$, *follows* $P(A) \leq P(B)$, (*monotony*)

e) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$, (*law of additivity*, *Additionsgesetz*)

f) $P\left(\bigcup_{i=1}^{n} A_i\right) \leq \sum_{i=1}^{n} P(A_i)$ (*subadditivity*, *Subadditivität*).

g) $P\left(\sum_{i=1}^{n} A_i\right) = \sum_{i=1}^{n} P(A_i)$, *for* $A_1, \ldots, A_n$ *pairwise disjunct* (*finite additivity*; *endliche Additivität*)

Proof of the Calculation Rules

a) Proof using K. axiom ii) and iii), with $A = 0$ and $B = \Omega$

b) and c) Proof follows from axiom i):

$$1 = P(\Omega) \qquad (\textit{from ii}))$$

$$1 = P(A + \bar{A})$$

$$1 = P(A) + P(\bar{A}) \qquad (\textit{from iii}))$$

d) Proof from:

$$B = A + (B \setminus A) \qquad (\textit{in junction with i}) \textit{ and ii}))$$

e) Proof by decomposing the set $A \cup B$,

in its disjoint subsets: $A \setminus B,\ A \cap B,\ \textit{and}\ B \setminus A$

with g) proven we get: $P(A \cup B) = P(A \setminus B) + P(A \cap B) + P(B \setminus A)$

with: $P(A) = P(A \cap B) + P(A \setminus B), \quad \textit{because of}\ A = A \cap B + A \setminus B$

$P(B) = P(A \cap B) + P(B \setminus A), \quad \textit{because of}\ B = B \cap A + B \setminus A.$

we can solve these two equations to:

$P(A \setminus B),\ \textit{and}\ P(B \setminus A),\ \textit{respectively and place the solution in}\ i).$

f) Proof from: $P(A \cup B) \leq P(A) + P(B)$

by full induction through n.

g) Proof follows from full induction of axiom iii) QED

Probability function

A ***probability function p***, is a function:

$$P : \Omega \rightarrow [0,1], \quad \textit{for which}:$$

i) $p(w) \geq 0$

ii) $p(\Omega) = 1$

iii) $1 = \sum_{w \in \Omega} p(w)$

holds.

Probability measure

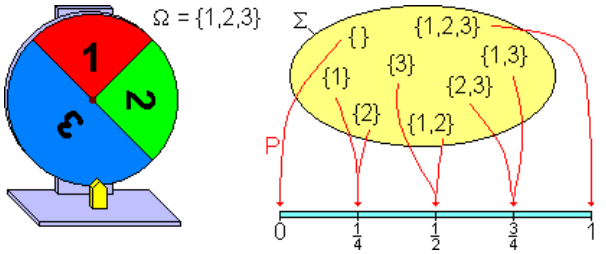
With Ω as a finite or countable infinite set, a function P from P(Ω) to [0,1] is called a ***probability measure*** for P(Ω), if:

$i)\ 0 \le P(A) \le 1, \quad \text{for all } A \in \Omega$ — Nichtnegativität, non-negativity

$ii)\ P(\Omega) = 1$ — Normiertheit, normalization

$iii)\ P\left(\bigcup_{i=1}^{\infty} A_i\right) \le \sum_{i=1}^{\infty} P(A_i),$ — Subadditivität, sub-additivity

$\text{for a series of } A_i \left(A_i \in \Omega,\ i = 1,2,\ldots\right) \text{ fulfilling: } A_i \cap A_j = 0, \text{ for } i \ne j \text{ holds.}$



Remark to the Definitions

Why is there a *probability function* p and a *probability measure* P?

Note: only P(A) is called the *probability* of A !

However, there is a strong interdependency between the *probability function* and the *probability measure.*

Let p be a probability function in Ω, and $A \subset \Omega$.

$P(A) := \sum_{w \in A} p(w)$ defines a probability measure.

Let P be a probability measure. Using $p : \Omega \to [0,1]$,

$p(w) := P(\{w\})$ defines a probability function.

Note:

The mystery lies in the *double brackets* that in this case mean that the *set* contains only one *element w*. The curse is to makes the distinction between sets and elements of the set! P(A) is the measure on the set A. So to indicate that it is a set that contains only *w* it is written ({*w*}) whiles on the left-hand side, it is an element so only one bracket.

Laplace probability

If we throw a die what is the likelihood of an uneven result?

For an honest die:

$$P(\{w\}) = \frac{1}{6}, \quad 1 \le w \le 6$$

$$P(\{1,3,5\}) = P(\{1\}) + P(\{3\}) + P(\{5\}) = 3\frac{1}{6} = \frac{1}{2}$$

Laplace definition of probability: $P(A) := \frac{|w|}{|\Omega|}$

with A as a subset of Ω



Pierre Simon Laplace (1749-1827)

Laplace probability

Say we throw the die two times and we got: (1,3). What is the probability of this result?

The set of all possible outcomes is:

$$\Omega = \{(1,1),(1,2),...,(6,6)\} = \{(i,j): 1 \le i \le 6, 1 \le j \le 6\},$$

$$|\Omega| = 36,$$

For $A_{(1,3)} \subset \Omega$, *we can define* :

$$P(A_{(1,3)}) := \frac{|A|}{|\Omega|} = \frac{1}{36}.$$

Laplace probability

Say we throw the die two times and we like to know the probability that the sum of the numbers is ≤ 9?

$$
\begin{aligned}
P(sum \le 9) = \quad & P(sum > 9) \\
= 1 - & P(sum \ge 10) \\
= 1 - & P\left(\overline{B_9}\right) \\
= 1 - & \frac{\left|\overline{B_9}\right|}{|\Omega|} \\
= 1 - & \frac{6}{36} = \frac{10}{12} = \frac{5}{6}
\end{aligned}
$$

$$
B_9 = \{(1,1),(1,2),\ldots,(4,5)\} \text{ and } \overline{B_9} = \{(4,6),(5,5,),(5,6),(6,4),(6,5),(6,6)\}
$$

Laplace Distribution

In a finite series of experiments with *n* trials, let the set be:

$$\Omega = \{w_1, w_2, ..., w_n\}, \quad with \quad n = |\Omega|,$$

for the individual probability of each elementary (probability function) event we can say:

$$p(w) = \frac{1}{n} = \frac{1}{|\Omega|}, \qquad w \in \Omega,$$

for a defined subset A (probability measure) we thus can say:

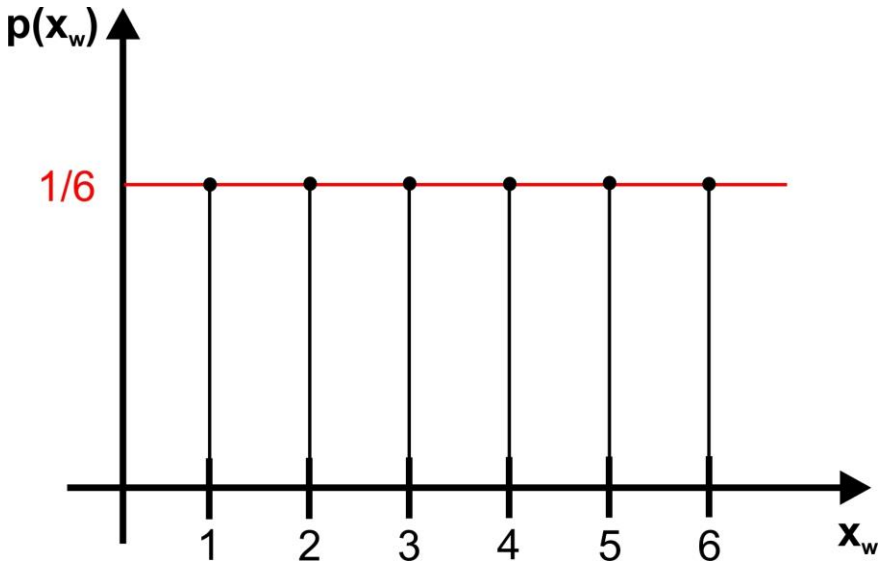
$$P(A) = \frac{|A|}{n} = \frac{|A|}{|\Omega|}, \qquad with\ A \subset \Omega\ .$$

In this case P is called: ***discrete uniform distribution***.

Such a distribution is called: ***Laplace distribution.***

Discrete Uniform Distribution

For every number of an honest die the probability in a single toss is equal.



Since every elementary event {*w*} has the same probability assigned p(*w*).

The probabilities are equally spaced.

Laplace Distribution, Example 1

Flipping an honest coin three times. We “define” head to be 0, and tail to be 1.
Our set is therefore going to be:

$$\Omega = \{0,1\}\ x\ \{0,1\}\ x\ \{0,1\}$$
$$\Omega = \{(0,0,0),(1,0,0),(0,1,0),(0,0,1),(1,1,0),(1,0,1),(0,1,1),(1,1,1)\}.$$

Our set Ω includes $|\Omega| = 2^3 = 8$.

In general: $|\Omega| = A^B$, with the base A = number of elements/trial
and the exponent B = number of trials

Laplace Distribution, Example 2

We continue flipping our coin (three times) but are now interested only in events that contain at least one head and one tail.

$$A = \{(1,0,0),(0,1,0),(0,0,1),(1,1,0),(1,0,1),(0,1,1)\}.$$

With |A| = 6,

we can calculate the probability

$$P(A) = \frac{|A|}{|\Omega|} = \frac{6}{8} = \frac{3}{4}.$$

Laplace Distribution, Example 3

What is the probability of the complementary of our example 2, i.e. “no head or no tail”?

$$\overline{A} = \{(0,0,0),(1,1,1)\}$$

$$P(\overline{A}) = \frac{|\overline{A}|}{|\Omega|} = \frac{2}{8} = \frac{1}{4}.$$

Of course, this result was easier to get...

Laplace Distribution, Example 4

Now for a more general question:

What would be the probability to get only one tail if we continue to flip the coin...?

Note: we stop flipping once we have a one

Our subset A will be:

$$A = \{(1),(0,1),(0,0,1),...\}$$

Now we can describe the probability as:

$$P(A) = P((1)) + P((0,1)) + P((0,0,1)) + ...$$

$$P(A) = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + ...$$

$$P(A) = 2^{-1} + 2^{-2} + 2^{-3} + ...$$

Since we continue to flip the coin infinite times we can say :

$$\sum_{i=1}^{\infty} 2^{-i} = 1 .$$

The Profit-checker... (fair)

How do I know whether a game is an “honest” one or not?

Scenario one: We toss the die again. This time we set money on the numbers.

Say we can choose one number (e.g. a “6”) and the input is 1 €.

The return is set as follows:

if the number is a “6” we get 5 € return (+ 1 € input).

if the number is not a “6”, we get 0 €.

Now we have to look for the average return (ar):

$$\mathrm{ar} = \mathrm{P}(6)\cdot \mathrm{return} - \mathrm{P}(\bar{6})\cdot \mathrm{input}$$

$$\mathrm{ar} = \frac{1}{6}\cdot 5 - \frac{5}{6}\cdot 1 = 0 \quad .$$

If the average return turns out to be 0, the game is considered a “fair” one.

The Profit-checker... (unfair)



Now we go to the Casino to play Roulette.

The conditions are as follows:

- We bet on a color (e.g. red) and the input has to be 100 €
- The game consists of 36 numbers and a “green” zero
- If you win you get 103 €

Now we have to look for the average return (ar) again:

$$ar = P(red) \cdot return - P(\overline{red}) \cdot input$$

$$ar = \frac{18}{37} \cdot 103 - \frac{19}{37} \cdot 100 = \frac{1854}{37} - \frac{1900}{37} = -\frac{46}{37} < 0$$

The average return is below 0, thus with ongoing trials the player will loose on average 1.2 €/round to the bank.

The Odds

If A is an event and Ā the complementary event:

$$R(A) := \frac{P(A)}{P(\overline{A})} = \frac{P(A)}{1 - P(A)}$$

is called the ***odds*** of A.

We can also define the **ratio of odds** for e.g. two events:

$$R(A:B) := \frac{R(A)}{R(B)} = \frac{\frac{P(A)}{P(\overline{A})}}{\frac{P(B)}{P(\overline{B})}} = \frac{\frac{P(A)}{1 - P(A)}}{\frac{P(B)}{1 - P(B)}} .$$

The Odds, Die Example 1

What would be the odds to toss a “6”, each time it’s our turn in a game of LUDO?

$$P(6) = \frac{|A|}{|\Omega|} = \frac{1}{6}$$

$$P(\bar{6}) = \frac{|\bar{A}|}{|\Omega|} = \frac{5}{6}$$



$$R(6) = \frac{P(6)}{P(\bar{6})} = \frac{P(6)}{1\text{-}P(6)} = \frac{1/6}{5/6} = \frac{1}{5} \ .$$

Every time the odds R(6) are 1 to 5.

The Odds, Die Example 2

How about the ratio of odds to toss a “6” vs. the odds to toss a “1” or a “2”?

Let A = {6} and B = {1,2},



$$P(6) = \frac{|A|}{|\Omega|} = \frac{1}{6} \qquad P(\bar{6}) = \frac{|\bar{A}|}{|\Omega|} = \frac{5}{6}$$

$$P(1,2) = \frac{|B|}{|\Omega|} = \frac{2}{6} \qquad P(\overline{1,2}) = \frac{|\bar{B}|}{|\Omega|} = \frac{4}{6}$$

$$R(1,2) = \frac{P(1,2)}{P(\overline{1,2})} = \frac{P(1,2)}{1-P(1,2)} = \frac{2/6}{4/6} = \frac{1}{2}$$

$$R(6) = \frac{1}{5}$$

the ratio of odds in this case is: $R(6:1,2) = \frac{R(6)}{R(1,2)} = \frac{2}{5}$.

The Odds, real world scenario

How about the ratio of odds to get killed in the US vs in Germany (2017)?



17284 murders in US (P(A)) with 325M citizens, 405 murders in Germany (P(B)) with 82M citizens,

$$P(A) = \frac{|A|}{|\Omega|} = \frac{17284}{325 * 10^6} \qquad P(\bar{A}) = \frac{|\bar{A}|}{|\Omega|} = \frac{325 * 10^6 - 17284}{325 * 10^6}$$

$$P(B) = \frac{|B|}{|\Omega|} = \frac{405}{82 * 10^6} \qquad P(\bar{B}) = \frac{|\bar{B}|}{|\Omega|} = \frac{82 * 10^6 - 405}{82 * 10^6}$$

$$R(A) = \frac{P(A)}{P(\bar{A})} = \frac{5.318 * 10^{-5}}{0.999947} = 5.318 * 10^{-5}$$

$$R(B) = \frac{P(B)}{P(\bar{B})} = \frac{4.939 * 10^{-6}}{0.999995} = 4.939 * 10^{-6}$$

the ratio of odds in this case is: $R(A:B) = \frac{R(A)}{R(B)} = \frac{5.318 * 10^{-5}}{4.939 * 10^{-6}} = 10.77$.

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Combinatorics

If we know that a Laplace model is appropriate for a given situation we can describe the probability of event A as the ratio:

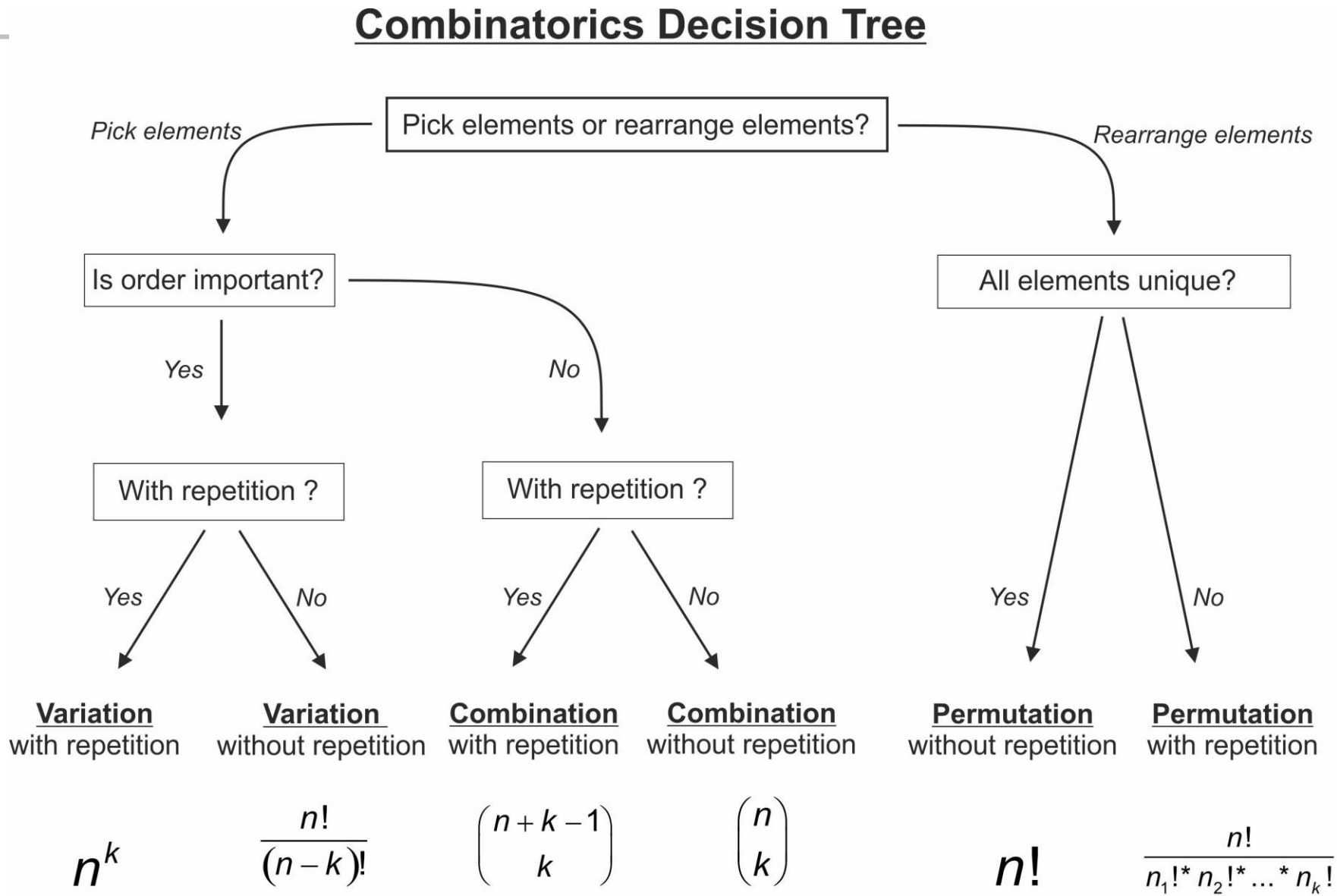
$$P(A) = \frac{number\ of\ events\ /\ elements\ of\ A}{number\ of\ all\ possible\ events}$$

In the end, probability is done counting wanted cases of A and possible cases => fundamental rule of counting...i.e.: *combinatorics*

We have to consider four different scenarios as a starting point:

- is the sample we want to take from the N (set):
  - a) in order?
  - b) not in order?
- is the sample placed back into the set or it is removed?:
  - a) placed back => multiple drawing are possible
  - b) not placed back => multiple drawing are impossible

Combinatorics



Ordered Sample with Replacement

In this model, every event *w* can be described by the k-tuple of the number of the chosen element in the row of occurrence. The set can therefore be described as:

$$\Omega_1 = \left\{ \left(w_1, ..., w_k \right) : 1 \leq w_i \leq n \right\}.$$

Every coordinate of the k-tuple can become n, and since all combination are possible (with replacement) we get:

$$\left| \Omega_1 \right| = \underbrace{(n \cdot n \cdot ... \cdot n)}_{k \text{ times}} = n^k.$$

Example is throwing a coin three times:

$$\Omega = \{0,1\} \; x \; \{0,1\} \; x \; \{0,1\}$$

$$\Omega = \{(0,0,0),(1,0,0),(0,1,0),(0,0,1),(1,1,0),(1,0,1),(0,1,1),(1,1,1)\}.$$

Our set Ω includes $|\Omega| = n^k = 2^3 = 8$.

Ordered Sample without Replacement

Again, in this model every event *w* can be described by the k-tuple. However, this time every element can occur only one time.

The set is therefore:

$$\Omega_2 = \left\{ \left(w_1,...,w_k \right) : 1 \le w_i \le n,\ w_i \ne w_j \ \text{for}\ 1 \le i \ \text{and}\ j \le n \right\}.$$

Again, the number of samples is identical to the number of k-tupels with the exception that:

$$\left|\Omega_2\right| = n \cdot (n-1) \cdot ... \cdot (n-k+1)$$

$$\left|\Omega_2\right| = \prod_{i=0}^{k-1} (n-i).$$

k ≤ n, because for k = n every sample is an ordering of the elements of the set Ω_2.

$$n \cdot (n-1) \cdot ... \cdot (n-k+1) \ \text{is but a permutation of}\ \Omega \ \text{and can be abbreviated}: n!$$

We get:

$$\left|\Omega_2\right| = n \cdot (n-1) \cdot ... \cdot (n-k+1) = \frac{n!}{(n-k)!}.$$

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Unordered Sample without Replacement

Once more, the number of samples is identical with the subsets of A with exactly k elements.

We get: $$\Omega_3 = \left\{ (A) \subset \{1,...,n\} \; : \; |A| = k \right\}.$$

Every element of Ω_3 is a combination of k out of n. $|\Omega_3|$ is therefore the sum of all combinations of k out of n.

$$|\Omega_3| = \frac{n \cdot (n-1) \cdot \cdots \cdot (n-k+1)}{k(k-1) \cdot \cdots \cdot 1}$$

$$\left|\Omega_3\right| = \frac{n!}{k! \cdot (n-k)!}.$$

This is defined as the ***binomial coefficient***: “n choose k”, or “n over k”

$$\frac{n!}{k! \cdot (n-k)!} = \binom{n}{k} = C_k^n.$$

The binomial coefficient provides the number of combinations one gets choosing k objects from a set of n objects, without replacement and ignoring the order of the objects.

Example for Unordered Sample without Replacement

(Lottery 6 out of 49). 6 balls are selected out of 49, which are labeled by their number. The ordering is neglected. It is ensured that all combinations (= unordered samples without replacements!) are equally likely. Thus, we have to deal with a uniform distribution.

Our set therefore is:

$$\Omega_3 = \left\{ (w_1, w_2, w_3, w_4, w_5, w_6) : 1 \le w_1 < w_2 < w_3 < w_4 < w_5 < w_6 \le 49 \right\}.$$

The number of possible combination of 6 out of 49 is:

$$\binom{n}{k} = \frac{n!}{k! \cdot (n-k)!} = \frac{49!}{6! \cdot (49-6)!} = 13983816 \quad .$$



The probability to have the “lucky“ 6 is:

$$P(A) = \frac{|A|}{|\Omega|} = \frac{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{49 \cdot 48 \cdot 47 \cdot 46 \cdot 45 \cdot 44} = \frac{1}{13983816} = 7.15 \cdot 10^{-8}.$$

Combinatorics Summary

For k-times choosing an object from n objects we can define:

	Choose with replacement	Choose without replacement
ordered	$\lvert\Omega_1\rvert = n^k$	$\lvert\Omega_2\rvert = (n)_k = \frac{n!}{(n-k)!}$
unordered	$\lvert\Omega_4\rvert = \binom{k+n-1}{k}$	$\lvert\Omega_3\rvert = \binom{n}{k}$